Xezae Peshlakai

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EDUCATION

University of California, Santa Cruz — Santa Cruz, CA
Bachelor of Science in Electrical Engineering | Expected Graduation: 2026 | GPA: 3.54
College Scholars Program Recipient
Relevant Coursework: Embedded Systems, Sensor Technologies, Electronics, Power Systems, Signal Processing, Control Systems, Electrical Infrastructure, Advanced Renewable Energy Sources, Storage, and Smart Grids

EXPERIENCE

ArchKey Solutions / Sprig Electric — Electrical Engineering Intern

San Jose, CA | June 2025 – August 2025

- Supported electrical engineering and construction teams on large-scale projects including Sutter Health Hospital, UCSC Kresge College, Google Campus, and 1X Technologies facilities.
- Assisted with electrical layouts, power distribution planning, and system coordination using AutoCAD, Revit, single-line diagrams, NEC standards, and California Title 24 requirements.
- Performed on-site verification of transformers, circuit breakers, and electrical installations to help ensure systems met project specifications and code requirements.
- Worked with engineers and field teams to review documentation, coordinate electrical systems, and support construction planning throughout different project phases.

Tech4Good — Web Developer

Santa Cruz, CA | September 2022 – March 2023

- Developed responsive front-end components using Angular, TypeScript, HTML, and CSS by turning Figma designs into functional web interfaces.
- Worked with cross-functional teams to improve usability, maintain consistent interface design, and support software development for research-focused platforms.
- Assisted with debugging and reusable component development to improve maintainability and development efficiency.

Rip Curl — Supervisor

Santa Cruz, CA | May 2022 – December 2024

- Led daily store operations and coordinated team responsibilities in a fast-paced environment, developing strong communication, leadership, and problem-solving skills through team management and customer interaction.
- Built long-term relationships with customers while helping maintain a community-oriented retail environment centered around surfing, outdoor culture, and customer experience.
- Assisted with scheduling, operational coordination, and resolving day-to-day issues while balancing team efficiency and customer service standards.

PROJECTS

Automated Solar Panel Cleaning System — Power and Mechanical Lead
Capstone Project | October 2025 – Present

- Designed and built a $600 autonomous solar panel cleaning robot for the residential market, aimed at homeowners who cannot easily access their roofs or prefer to avoid recurring third-party cleaning services.
- Integrated distributed 24V/12V/5V power systems, embedded monitoring, relay protection, actuator control, and automated fault detection into a fully autonomous cleaning platform.
- Developed embedded monitoring and protection systems using STM32 microcontrollers, sensors, ADCs, and relay control to detect abnormal electrical conditions and safely shut down hardware during operation and testing.
- Designed electromechanical assemblies and electrical schematics using Onshape and KiCad, creating custom 3D-printed mounting hardware, motor driver interfaces, DC-DC converter layouts, and voltage/current monitoring systems.
- Completed 6+ rounds of hardware validation using oscilloscopes, multimeters, MATLAB analysis, serial diagnostics, and iterative subsystem testing to improve system reliability and monitoring performance.

SKILLS

Embedded Systems: STM32, PIC Microcontrollers, GPIO, ADC, PWM, Sensor Integration, Embedded C

Power & Electrical: Power Distribution, Relay Protection, DC-DC Converters, Voltage/Current Monitoring, NEC Standards, Title 24

Hardware & CAD: KiCad, Onshape, AutoCAD, Revit, Electrical Schematics, Electromechanical Design, 3D Printing

Software & Analysis: MATLAB, Python, HTML/CSS, TypeScript, Angular, Figma

Engineering Tools: Oscilloscopes, Multimeters, Hardware Testing, Troubleshooting, System Validation, Gantt Charts, Cross-Functional Collaboration